

Assignment 5 CSE 16-02: Applied Discrete Mathematics Winter 2026

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Complete the questions carefully and show your work. Please remember to:

- Scan the pages and combine them into a single PDF file. Make sure your work is readable and that pages are oriented correctly.
- Upload your PDF file to Canvas/Gradescope under Assignment 5.
- In Gradescope select the page(s) for each problem.

Notation: $\emptyset$ is the empty set, $\mathbb{N}$ is the set of natural numbers, $\mathbb{Z}$ is the set of integers, $\mathbb{Q}$ is the set of rational numbers, and $\mathbb{R}$ is the set of real numbers. $\mathcal{P}(A)$ is the power set of A .

[46 points total in assignment.]

0. Each answer is clearly written and each answer is correctly mapped to its rubric on Gradescope. [4 points]
1. For $a, b \in \mathbb{N}$, prove the following statements below. Hint: to prove that g is the $\gcd(a, b)$ you need to prove that $g|a$ and $g|b$ and that g is the greatest such common divisor. [10 points]
 - (a) Let $a = 5x$ and $b = 5y + 7$ for some x and y in $\mathbb{N}$. Prove that $\gcd(a, b) = \gcd(x, b)$. Write a proof that shows: (i) that $\gcd(a, b)|x$, then (ii) show that it implies that $\gcd(a, b)$ equals $\gcd(x, b)$. [2 points]
 - (b) If both a and b are divisible by 4, then $\gcd(a, b) = 4 \gcd(a/4, b/4)$. [2 points]
 - (c) If m is a common divisor of a and b , show the general case of the previous question, namely that $\gcd(a, b) = m \gcd(a/m, b/m)$. [2 points]
 - (d) Prove that $\gcd(a, b) = \gcd(a + b, b)$. [2 points]
 - (e) Let a and b be integers, not both zero. Prove that $\gcd(a, b)$ is the smallest positive integer that can be written in the form $ax + by$ for integers x and y .

Sol

- (a)
 - i. Since $5y + 7$ is not divisible by 5, the $\gcd$ of a and b must divide $a = 5x$, by Euclid's Lemma, the $\gcd$ must divide x .
 - ii. **Forward Direction:** Suppose d is a common divisor of a and b ($d|a$ and $d|b$). From part (i), we already proved that if d divides both $5x$ and b , then $d|x$. Thus, d is a common divisor of x and b . **Reverse Direction:** Suppose d is a common divisor of x and b ($d|x$ and $d|b$). If $d|x$, then d must also divide any multiple of x , specifically $d|5x$. Since $a = 5x$, d divides a . Thus, d is a common divisor of a and b .
- (b) Since a and b are divisible by 4, there exists a', b' such that $a = 4a', b = 4b'$. We want to show there exists some g' such that $g = \gcd(a, b) = 4g'$. Let d be the $\gcd$ of a' and b' . We then know that $4d|a$ and $4d|b$. Since 4 is a common factor of a and b , the $\gcd$ of a and b can be written as $4c'$. Chaining everything together we get

$$\gcd(a, b) = 4c' = 4d = 4 \gcd(a', b') = 4 \gcd\left(\frac{a}{4}, \frac{b}{4}\right) \quad (1)$$

- (c) The proof is identical but replace 4 with m .

- (d) $(\leq)$

Let $x = \gcd(a, b)$. Since x divides a and b , it must also divide $a + b$ (note $a + b = x(a' + b')$ for some a', b' by definition of divides). This means that $\gcd(a + b, b)$ must be at least $\gcd(a, b)$.

$(\geq)$

Let $y = \gcd(a + b, b)$. y must also divide $a + b - b$ (let $\alpha_1 y = a + b$ and $\alpha_2 y = b$. This means $(\alpha_1 - \alpha_2)y = a + b - b$). This means that $\gcd(a, b)$ must be at least $\gcd(a + b, b)$.

Since they are $\geq$ and $\leq$, they must be equal.

- (e) Consider the set of all positive numbers that $ax + by$ form given that a, b are nonzero and x, y are in the integers. This set is nonempty since at least one of the two a, b is nonzero. By the Well Ordering Principle, there must exist a least element. Call this least element d . By the division algorithm, we know that

$$\begin{aligned} a &= dq_a + r_a \\ b &= dq_b + r_b \end{aligned} \tag{2}$$

for some $q_a, q_b \in \mathbb{Z}$ and $r_a, r_b \in \{0, \dots, d-1\}$. We will work with $r = r_a, q = q_a$ and the proof proceeds without loss of generality with b .

$$r = a - dq$$

$$r = a - (ax + by)q$$

$$r = a(1 - xq) + b(-yq)$$

This shows that r is also a linear combination of a and b . If $r > 0$, then r would be an element of S . However, we stated that d is the smallest positive element of S , and the division algorithm tells us $r < d$. The only way to avoid a contradiction is if $r = 0$. Therefore, $d \mid a$. By the same logic, $d \mid b$. Thus, d is a common divisor of a and b .

2. We will prove that if n is an odd integer $\in \mathbb{N}$, then 16 divides $n^4 - 1$. We're going to do this in two ways. **[7 points]**

- First, prove that if n is an odd integer $\in \mathbb{N}$, then 8 divides $n^2 - 1$.
- Prove that if n is an odd integer, then $n^2 + 1$ is even.
- Using the identity $n^4 - 1 = (n^2 - 1)(n^2 + 1)$ and your results from steps 1 and 2, prove that 16 divides $n^4 - 1$.
- Let $n = 2k + 1$. Express $n^4 - 1$ using the binomial theorem as a sum of terms of the form $a_i k^i$. Write this sum explicitly.
- Consider the last two terms of your sum, $a_2 k^2$ and $a_1 k$. Rewrite $a_2 k^2 + a_1 k$ in the form $bk^2 + c(k^2 + k)$ where b is an integer.
- Show that b and $c(k^2 + k)$ is divisible by 8.
- Prove the result stated at the beginning of this Problem, using the previous three steps.

sol

- (a) Since n is odd, we can represent it in the form:

$$n = 2k + 1$$

where k is any integer. Substitute $(2k + 1)$ for n in the expression $n^2 - 1$:

$$(2k + 1)^2 - 1 = 4k^2 + 4k = 4k(k + 1)$$

by law of excluded middle, k or $k+1$ must be even. the product of any number with an even number must be even, thus $k(k + 1)$ must be even. Therefore $k(k + 1)$ can be written as $2k_0 = k(k + 1)$ for some k_0 . Together this means $n^2 - 1 = 8k_0$ which means it is divisible by 8.

- (b) if n is odd, it can be written as $2k + 1$.

$$(2k + 1)^2 + 1 = 4k^2 + 4k + 2 \tag{3}$$

this number is clearly divisible by 2 and is thus even.

- (c) since $n^2 - 1 = 8k_0$ and $n^2 + 1 = 2k_1$ for some k_0, k_1 in $\mathbb{Z}$, the product can be written as $16k_0k_1$ and thus

$$n^4 - 1 = (n^2 - 1)(n^2 + 1) = 16k_0k_1 \tag{4}$$

is divisible by 16.

(d) Let $n = 2k + 1$. We expand $(2k + 1)^4$ using the binomial theorem:

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}$$

For $n^4 = (2k + 1)^4$:

$$(2k)^4 + 4(2k)^3 + 6(2k)^2 + 4(2k)^1 + 1^4 = 16k^4 + 32k^3 + 24k^2 + 8k + 1 \quad (5)$$

Now, we subtract 1 to get $n^4 - 1$:

$$n^4 - 1 = 16k^4 + 32k^3 + 24k^2 + 8k$$

(e) We focus on the terms $a_2k^2 + a_1k$, which are $24k^2 + 8k$. We want to rewrite this in the form $bk^2 + c(k^2 + k)$. Notice that $24k^2$ can be split into $16k^2 + 8k^2$.

$$\begin{aligned} 24k^2 + 8k &= 16k^2 + 8k^2 + 8k \\ &= 16k^2 + 8(k^2 + k) \end{aligned} \quad (6)$$

Here, $b = 16$ and $c = 8$.

(f) Since both terms are divisible by 8, the whole term is divisible by 8.

(g)

$$\begin{aligned} 16k^4 + 32k^3 + 24k^2 + 8k &= 16k^4 + 32k^3 + 16k^2 + 8k^2 + 8k \\ &= 16(k^4 + 2k^3 + k^2) + 8(k^2 + k) \\ &= 16(k^4 + 2k^3 + k^2) + 8(k + 1)(k) \end{aligned} \quad (7)$$

since $(k + 1)(k)$ is even, both terms are divisible by 16 and thus the original expression $n^4 - 1$ is divisible by 16.

3. Prove or disprove: $\forall x \in \mathbb{R}, \sqrt{x} \notin \mathbb{Q} \rightarrow x \notin \mathbb{Q}$. [5 points]

Sol

This statement is false. A counterexample is $\sqrt{2}$ which is not in $\mathbb{Q}$, but 2 is.

4. Prove or disprove: $\sqrt[3]{3} \in \mathbb{Q}$. [5 points]

Sol

This statement is False.

Assume, for the sake of contradiction, that $\sqrt[3]{3}$ is a rational number. By definition, there exist two integers p and q ($q \neq 0$) such that:

$$\sqrt[3]{3} = \frac{p}{q}$$

Assume that $\frac{p}{q}$ is in its simplest form, i.e. $\gcd(p, q) = 1$.

$$\begin{aligned} \left(\sqrt[3]{3}\right)^3 &= \left(\frac{p}{q}\right)^3 \\ 3 &= \frac{p^3}{q^3} \end{aligned} \quad (8)$$

from this, we see that p^3 is divisible by 3. Since 3 is prime, p must be divisible by 3 by Euler's Theorem. We can rewrite p as $3p_1$.

$$(3p_1)^3 = 3q^3 = 27p_1^3 \quad (9)$$

which implies

$$q^3 = 9p_1^3 = 3(3p_1^3) \quad (10)$$

This also means that q^3 is divisible by 3, which applying Euler's Theorem we see implies q is also divisible by 3.

q and p are divisible by 3 which is a contradiction since we assumed $\gcd(q, p) = 1$.

5. Prove: If $n, m \in \mathbb{Z}$ and $7n^4 + 11m^3 = 214$ then n and m have the same parity. [5 points]

Sol

Let us take the mod 2 of both sides

$$\begin{aligned} (7n^4 + 11m^3) \mod 2 &= (214) \mod 2 \\ (n^4 + m^3) \mod 2 &= (0) \mod 2 \end{aligned} \tag{11}$$

One can check exhausting all combinations of parity for n, m , but a proof by contradiction is simpler. Assume for the sake of contradiction that n, m have opposite parity. By induction, taking any number to some positive power retains its parity. Adding two numbers with opposite parity always results in parity 1 which is a contradiction.

6. Prove or disprove: If $x \notin \mathbb{Q}$, $y \in \mathbb{R}$, $x \neq 0$, and $y \neq 0$, then for all $n, m \in \mathbb{N}$, $x^n y^m \notin \mathbb{N}$. [5 points]

Sol

This statement is Incorrect. A counterexample lies in $x = \sqrt{2}, y = 1, n = 2, m = 1$. (note, to prove that the forall is incorrect, we only need to find a single instance that fails the claim).

$$\sqrt{2}^2 \cdot 1^1 = 2 \in \mathbb{N} \tag{12}$$

7. Consider the sets

$$S_1 = \{k^2 + 5 : k \text{ is an even natural number}\}, S_2 = \{4k + 5 : k \in \mathbb{N}\}.$$

- (a) Prove $S_1 \subseteq S_2$. [3 points]
 (b) Prove $\neg(S_2 \subseteq S_1)$. [2 points]

Sol

- (a) Let s_1 be an arbitrary element in S_1 . Since k is even, it can be written as $2k_0$ for some $k_0 \in \mathbb{Z}$.

$$k^2 + 5 = (2k_0)^2 + 5 = 4k_0^2 + 5 \tag{13}$$

For any k_0 in $\mathbb{Z}$, the square of that number is a nonnegative integer, i.e. in $\mathbb{N}$. The form of this number matches with the specification in S_2 , thus it is an element in S_2 . Thus all elements in S_1 are elements in S_2 , i.e. $S_1 \subseteq S_2$.

- (b) To prove that S_2 is not a subset of S_1 , we only need to find a single element in S_2 that is not in S_1 . Consider

$$4(2) + 5 \in S_2 \tag{14}$$

this element is not in S_1 since no even natural number squares to 8. ($\sqrt{8} = 2\sqrt{2}$ is irrational.)